

HW3

Sunday, February 22, 2026

6:22 PM

Least Squares & MLE

model specification: $Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_j X_{ij} + \epsilon_i$, $i = 1, \dots, n$, $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$

~det joint PDF: $P((X,Y) \in \mathbb{R}) = \int_{\mathbb{R}} f_{X,Y}(x,y) dx dy$; likelihood eval. this density eval. at obs. values

~det given model specification: $Y_i \sim N(\mu_i, \sigma^2)$ w/ each obs: $f(y_i) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(y_i - \mu_i)^2}{2\sigma^2}\right)$

$\therefore$ the joint density of obs $f(y_1, \dots, y_n) = \prod_{i=1}^n f(y_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(y_i - X_i\beta)^2}{2\sigma^2}\right)$
 $\uparrow$
 $L(\beta, \sigma^2)$

$\rightarrow$ log likelihood: $l = \log \left[\prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(y_i - X_i\beta)^2}{2\sigma^2}\right) \right]$ using prop: $\log(\prod a_i) = \sum \log(a_i)$

$$= \sum_{i=1}^n \log \left(\frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(y_i - X_i\beta)^2}{2\sigma^2}\right) \right) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \underbrace{\sum_{i=1}^n (y_i - X_i\beta)^2}_{RSS}$$

$\rightarrow \frac{dl}{d\beta} = \frac{\text{first term not depend on } \beta}{\text{first term not depend on } \beta} - \frac{1}{2\sigma^2} \frac{\partial RSS}{\partial \beta} \stackrel{\text{set } 0}{=} 0 \rightarrow \frac{\partial RSS}{\partial \beta} = 0$

~Using Vector Rep: $Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \end{bmatrix}$, $X = \begin{bmatrix} X_1^T \\ \vdots \\ X_n^T \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & \dots & X_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \dots & X_{np} \end{bmatrix}$, $\beta = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_p \end{bmatrix}$; ex: $X\beta = \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 X_1 \\ \beta_0 + \beta_1 X_2 \\ \vdots \\ \beta_0 + \beta_1 X_3 \end{bmatrix}$

~residuals: $Y - X\beta = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \end{bmatrix}$; $RSS = e_1^2 + e_2^2 + \dots$ $\uparrow$ intercept e estimate of $\epsilon = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \end{bmatrix}$ $\therefore RSS = e^T e = (Y - X\beta)^T (Y - X\beta)$

Returning to derivation:

$$\frac{\partial RSS}{\partial \beta} = -2X^T(Y - X\beta) \stackrel{\text{set } 0}{=} 0 \rightarrow X^T(Y - X\beta) = 0 \rightarrow \beta = (X^T X)^{-1} X^T Y$$

which is the OLS estimator of β

Mallo C_p : given p-parameters $= \frac{RSS}{\sigma^2} - n + 2p$

AIC_(gaussian): given p-params $= 2p - 2\log(\hat{L})$

$\sim \ln(\hat{L}_{max}) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{RSS}{2\sigma^2}$
 $\uparrow$ max likelihood
 $\uparrow$ derived above

$\therefore AIC = 2p + \underbrace{n \log(2\pi\sigma^2)}_{\text{constant}} + \frac{RSS}{\sigma^2}$

From the derivation, AIC & C_p differ by a constant (the 'n' term in C_p), therefore since constants cancel during model selection, C_p & AIC are equiv. for gaussian linear regression.